XIAO CUI, ZHENG

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EDUCATION

- **Nanjing University** Sep 2020 – Jun 2026
  Ph.D. in Chemical Biology **|** *Supervisor*: Prof. Ran Xie
  Thesis: Precise Editing of Proteins and Glycans via Glycoenzyme Engineering
- **Northeast Forestry University** Sep 2016 – Jun 2020
  B.S. in Engineering Science **|** *Cumulative GPA*: 3.8/4.0

RESEARCH EXPERIENCE

A Genetically Encoded Photoactivatable Glycoenzyme Platform for Spatiotemporal Glycan Editing

- Engineered a genetic-code-expansion system to site-specifically incorporate a photocaged unnatural amino acid (ONBY) into glycoenzyme, achieving >80% activity recovery within 5 min UV irradiation.
- Achieved light-controlled glycan editing at single-cell resolution, and extended to upconversion nanoparticle-assisted deep-tissue editing in a mouse tumor model.
- Offers a general strategy for spatiotemporally controlled precise visualization and manipulation of cell-surface glycans and providing new tools for probing dynamic glycan regulation.

Traceless Protein-Selective Glycan Labeling and Chemical Modification

- Developed a dual-regulated (chemical inhibition + light-triggered release) editing platform using ligand-photocleavable linker-glycoenzyme complexes, enabling precise glycan modification on specific proteins without persistent enzyme retention.
- Demonstrated that prolonged glycoenzyme residence perturbs target protein turnover and dimerization, establishing the “edit-and-leave” principle for accurate functional interpretation of glycosylation.
- Provides a new chemical tool for ligand–enzyme conjugates and controlled modulation of immune checkpoints in cancer therapy.

PUBLICATIONS

[1] **Zheng, X.**; Wei, W.; Wang, C.; Du, Y.; Yang, J.; Zhou, Y.; Liu, Y.; Liang, Y.; Ding, L.; **Xie, R.*** A Photoactivatable Glycoenzyme Platform for Spatiotemporally Controlled, Protein-Selective, and In Vivo Glycan Editing. ***J. Am. Chem. Soc.*** 2026, 148 (10), 10990-11000.

[2] **Zheng, X.**; Li, Y.; Cui, T.; Yang, J.; Meng, X.; Wang, H.; Chen, L.; He, J.; Chen, N.; Meng, L.*; Ding, L.*; **Xie, R.*** Traceless Protein-Selective Glycan Labeling and Chemical Modification. ***J. Am. Chem. Soc.*** 2023, 145 (43), 23670-23680.

[3] **Zheng, X.**; He, B.; Meng, L.*; **Xie, R.*** Recent Advances in Protein-Specific Glycan Imaging and Editing. ***ChemBioChem*** 2023, 24 (9), e202200778.

[4] Yang, J.[#]; Wang, K.[#]; Zhang, S.; **Zheng, X.**; Cui, T.; Yang, X.; Liu, Y.; Lu, D.; Wang, Y.; Tian, X.; Zhang, P.; Huang, S.*; **Xie, R.*** Site-Specific Introduction of Bioorthogonal Handles to Nanopores by Genetic Code Expansion. ***Angew. Chem. Int. Ed*.** 2023, 62 (21), e202216115.

[5] Yao, Y.; Jia, R.; Liu, C.; Wang, H.; Li, T.; **Zheng, X.**; Zhong, T.; Feng, N.; Sun, J.; Li, K.; **Xie, R.**; Ding, L.; Yan, C.; Ding, L.*; Ju, H. An *In-Situ*-Tag-Generation Proximity Labeling Technology for Recording Cellular Interactions. ***Angew. Chem. Int. Ed*.** 2023, 63 (31), e202407109.

GRANTS

- National Natural Science Foundation of China Young Students Basic Research Project, Project Leader 2024-2025
- Nanjing University Doctoral Innovation Leadership Project, Project Leader 2025-2026

SELECTED CONFERENCES

- 74th Lindau Nobel Laureate Meeting (2025, Germany) — Selected as one of 30 PhD students nationwide for a two-week academic exchange.
- 2nd International Symposium on Chemical & Biomedical Imaging (2026, Hong Kong) — Poster presentation.
- 31st International Carbohydrate Symposium (2024, Shanghai) — Poster presentation.
- Tang Aoqing Doctoral Forum (2023, Changchun) — Oral presentation.
- Peking University Graduate Forum on Chemical Biology (2023, Beijing) — Oral presentation.
- 7th National Youth Science and Technology Forum on Glycoscience (2026, Shenyang) — Poster presentation.
- 12th National Conference on Chemical Biology (2023, Dalian) — Poster presentation.

SELECTED HONORS AND AWARDS

- **Outstanding graduates** 2026
  Awarded by Nanjing University.
- **Lindau Scholar** 2025
  Awarded by Govt. of China and the German National Science Foundation to doctoral students selected to attend the 74th Nobe Laureate Meeting.
- **National Scholarship** 2023
  Awarded by the Ministry of Education for achieving the best academic results.
- **Outstanding Postgraduate Student Pacesetter** 2023
  Awarded by Nanjing University for outstanding performance during postgraduate studies.
- **First Prize for Oral Presentation** 2023
  Awarded by Jilin University for outstanding performance in the oral presentation.
- **Best Poster Award** 2026
  Awarded by Nanjing University for excellent performance at the 6th Interdisciplinary Forum for Doctoral Students.
- **“Glycoscience & Therapy” Outstanding Poster Award** 2026
  Awarded by Northeast University for outstanding performance in the poster presentation.

TEACHING ASSISTANTSHIP

- Served as a teaching assistant for multiple undergraduate courses, including Fundamentals of Glycoscience, Critical Thinking, and Innovation Experiments for Undergraduates.
- Supervised undergraduate thesis projects and provided guidance on experimental design, data analysis, and scientific writing.

PERSONAL SKILLS

- **Experimental Skills:**

◇ Proficient in routine molecular and cellular biology techniques, including microbial/cell culture

(including primary cells), plasmid construction, mammalian cell line engineering, protein expression and purification, Western blotting, ELISA, flow cytometry, and fluorescence imaging.

- ◇ Skilled in the synthesis of unnatural amino acids, preparation of small-molecule probes, bioorthogonal chemistry, and chemical conjugation/labeling techniques.
- ◇ Experienced in operating various large-scale instruments, including confocal microscopes (Leica SP8, Dragonfly, Olympus), flow cytometry analyzers/cell sorters, AKTA, BLI, PCR, microplate readers, HPLC, electrospray quadrupole high-resolution mass spectrometers, and LC–MS.
- ◇ Proficient in data processing and visualization using software such as Imaris, ChemDraw, FlowJo, SnapGene, GraphPad Prism, Adobe Illustrator, ImageJ, and PyMOL.

- **Project Management:** Demonstrated capability in independent research project design and management, evidenced by the successful acquisition and completion of a competitive National Natural Science Foundation of China (NSFC) grant.
- **Languages:** English (IELTS: 6.5; CET-6: 546).